

Taming Unleashed Large Language Models With Blockchain for Massive Personalized Reliable Healthcare

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Abstract—The digital health field’s pursuit of massive, personalized healthcare continuously faces constraints from doctors’ resources and capacity limitations. Recently, the emergence of large language models (LLMs), with their remarkable comprehension and processing abilities, has revolutionized digital health and enhanced massive, personalized healthcare. Although these LLMs have achieved significant advancements, they have also introduced inevitable hallucinations, which impact patient safety when used in massive applications. To address these challenges, this study proposes a digital hospital for a massive, personalized, reliable healthcare service named the Chat Chain-Brain-based Doctor (CHATCBD). In addition, this study transforms the LLM-based diagnostic process into a digital hospital architecture, designs a controllable AI agents framework, and develops a self-audit mechanism to enhance their reliability. The proposed CHATCBD uses blockchain technology to decentralize external regulation of the LLMs’ personalized diagnoses. It introduces a blockchain-based personalized routing management mechanism to improve patient-centered decision-making and designs a blockchain-based audit framework based on a proposed mathematical model that ensures both the professionalism and honesty of audits, serving as a safety net for addressing LLM hallucinations. The results of extensive experiments conducted on 13 datasets from multiple perspectives demonstrate that the proposed CHATCBD system can significantly enhance the capabilities of LLMs in personalized healthcare.

Index Terms—Agent, blockchain, digital health, large language model, reliability.

I. INTRODUCTION

PERSONALIZED healthcare is a high-quality diagnosis requiring a patient-centered, customized analysis. Achieving massive personalized healthcare has always been a persistent goal in the healthcare field [1]. However, there is a global shortage of health workers, with estimates suggesting a shortfall of 15 million healthcare professionals by 2030 [2]. The limited availability of medical professionals makes it extremely challenging to provide timely medical expertise to every patient. However, with the advancement of digital health technologies, the integration of innovative digital and intelligent solutions, has become a pivotal strategy to meet the healthcare needs of large patient populations [3].

The recent integration of large language models (LLMs) into healthcare has marked a crucial shift, making artificial intelligence (AI) play a key role in medical diagnostics [4]. The capabilities of LLMs allow them to assimilate and interpret diverse data sources [5], which will assist in medical diagnostics [6]. However, recent research has highlighted several emerging challenges [7]. LLMs often generate misdiagnoses and biased results, whereas personalized medicine requires reliable step-by-step reasoning based on a patient’s unique physiological information and medical history. Therefore, if LLMs are used in personalized treatment, their hallucination problem could cause irreversible consequences [8].

The most common methods for mitigating hallucinations in LLMs in specialized scenarios include multi-agent internal collaboration reasoning and retrieval-augmented external knowledge base construction [9]. Multi-agent collaboration, which is achieved by establishing internal coordination through the meticulous design of prompts, aims to elicit domain-specific knowledge embedded in a model during the model training process [10]. However, in the field of personalized healthcare, due to the lack of customized patient data, LLMs cannot directly output personalized summaries and inferences [11]. Hence, relying only on internal coordination mechanisms can be inadequate [12]. Retrieval-augmented generation (RAG) is an external mechanism that enhances prompting, and its application in specific domains has proven it to be an important means of improving the accuracy of LLMs [13]. Personalized healthcare greatly relies on reliable patient medical histories that are often dispersed across various medical institutions,

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The code used in this study can be found at <https://github.com/LDDLQ/ChatCBD>.

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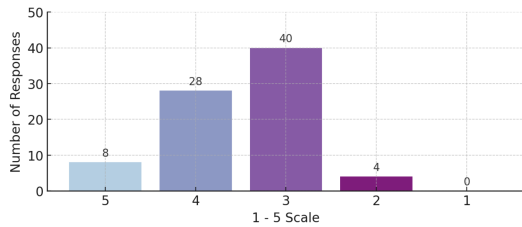


Fig. 1. Survey results: evaluating the GPT-4' reliability and consistency in healthcare using the five-point scale, where "0" indicates low and "5" stands for high).

clinics, and personal health records [14]. The LLMs can conduct step-by-step reasoning centered around a patient's medical history, thus generating more personalized plans. However, to use this data for RAG, more advanced measures need to be implemented.

Blockchain technology provides new opportunities for addressing the aforementioned challenges [15]. The application of blockchain to the healthcare domain has primarily focused on two areas: smart contracts in Ethereum for providing an efficient and tamper-proof solution for tracking patients' electronic health records (EHRs) histories [16] and smart contracts for constructing a transparent and auditable regulatory model [17]. Previous research has recognized blockchain as a trusted platform for personalized medical knowledge storage [18] and inference but also inspires us to use distributed intelligence to regulate the LLM output.

This study combines the LLMs with blockchain technology and establishes an innovative digital hospital geared toward reliable, massive, personalized healthcare service called Chat Chain-Brain-Based Doctor (CHATCBD). The CHATCBD transforms the diagnostic process based on LLMs into a more personalized, step-by-step reasoning process similar to that in hospitals and then turns this process into a blockchain-enhanced regulatory framework.

The CHATCBD is structured to emulate the process flow of a traditional hospital by segmenting its operations into four distinct systems: consulting, diagnosing, auditing, and documenting systems. Each of the four systems is tailored to personalized healthcare and meticulously designed to incorporate various medical roles facilitated by LLMs to interact effectively with patients. These interactions are seamlessly augmented and regulated using reliable, audited information obtained from the external blockchain, retrieving, augmenting, and concluding optimal solutions. The blockchain also allows supervision and feedback from a multitude of human experts [19]. The main contributions of this work can be summarized as follows:

- 1) A multi-agent architecture for personalized healthcare is proposed. The proposed architecture helps make model diagnostics controllable, and in the face of personalized needs, this study explores the applications of role-playing and an internal self-audit mechanism in healthcare multi-agent systems;
- 2) A blockchain-based personalized routing management mechanism is implemented into an RAG framework. This mechanism uses the blockchain-based EHRs of past patients and the diagnostic results of the CHATCBD.

The CHATCBD integrates conversational context with blockchain-based memory across the four systems, ensuring the reliability of personalized healthcare while reducing redundant blockchain-based audits;

- 3) A blockchain-based audit framework is developed to enhance the LLM performance by leveraging distributed intelligence for reliable human audits to mitigate hallucinations in personalized medical diagnoses obtained using LLMs. Grounded in Nash equilibrium, the proposed framework ensures fairness and integrity in expert evaluations and decision-making processes;
- 4) The effectiveness of the proposed CHATCBD's internal-external collaborative architecture in personalized diagnosis is demonstrated by extensive experiments on public and private datasets. The results show that integrating blockchain with LLMs can significantly enhance the reliability and consistency of diagnostics while increasing precision in personalized healthcare, with only a modest impact on the system's performance and cost.

The remainder of this paper is organized as follows. Section II provides an overview of the application progress of the LLM and blockchain technology in the healthcare field and outlines the inspiration behind this research. Section III explains the operational logic of the four systems in the proposed CHATCBD. Section IV introduces the operational algorithm of the proposed system. Section V presents the experiments and analysis. Finally, Section VI summarizes this study.

II. RELATED WORK

The personalized healthcare discussed in this study aims to acknowledge individual patient differences by optimizing medical services based on a patient's medical history rather than adopting a one-size-fits-all approach [20]. LLMs face certain challenges when addressing personalized healthcare reasoning tasks, including hallucinations, EHR provenance management, and the lack of regulation. To address these challenges, this chapter explores the potential of blockchain.

A. LLM Applications in Personalized Healthcare

Recently, LLMs like ChatGPT [21] and GPT-4 [22] have exceptional capabilities in handling complex language tasks [23]. However, when general-purpose models are applied to specific domains, their performance often decreases [24]. Particularly in the healthcare field, LLMs can generate misdiagnoses and exhibit biases [25].

Through a survey of 80 experienced GPT-4 users from Nanjing Drum Tower Hospital, this study has assessed their views on GPT-4 in real healthcare inquiries. Fig. 1 shows that only 10% ($n = 8$) of the users considered GPT-4 highly reliable and consistent and awarded it with a top score; however, 50% ($n = 40$) of users rated the LLMs' reliability and consistency as moderate with a score of three. Additionally, personalized healthcare requires high quality reasoning [26]. However, despite LLMs' reliance on extensive pre-trained datasets, they face limitations when handling highly personalized data [27]. Although the pre-trained datasets cover a range of knowledge,

they cannot fully represent the unique physiological conditions of every patient [28].

B. LLM Agents and RAG

Recent research has demonstrated that AI agent technologies can enhance LLM performance in handling complex tasks like CAME [29], AgentVerse [30], MetaGPT [31], ChatDev [32], and AutoGen [33]. In addition, LLM-driven agents have been applied to the decision-making, control, and other robotics-related fields, which has been evidenced by studies like March in Chat [34], ChatGPT for Robotics [35], Task Planning Agent [36], and Statler [37]. These studies show that designing multi-agent systems in specialized domains can enhance an LLM's performance. Meanwhile, recent research has also surprisingly witnessed the success of RAG as an external mechanism that can effectively collaborate with multi-agent systems [38]. Further, integrating RAG into multi-agent systems can significantly enhance problem-solving performance in specific domains, improving accuracy by combining information retrieval and language generation capabilities.

However, the existing research on the LLMs remains unsuitable for healthcare scenarios [39]. Unlike the fields of law and mathematics, the application of LLMs to healthcare involves critical matters of life and safety. The inherent hallucination flaws of LLMs [40] have led most studies to assume that more reliable regulatory measures are still needed [41]. Although RAG can effectively enhance the performance of LLMs, personalized healthcare greatly rely on a patient's historical data and physiological information for reasoning. Personal medical data are often dispersed across different platforms, which introduces various problems, such as data trust management and quality control [42]. This highlights the need for developing more reliable and controllable frameworks which could be a promising solution [43].

C. Blockchain Application to Healthcare

Blockchain technology has emerged as a promising solution to various challenges in the healthcare sector [44]. Blockchain is characterized by decentralization, transparency, immutability, and traceability [45]. In addition, the introduction of smart contracts allows the execution of various general-purpose program models on-chain, which enables automated and tamper-resistant protocol execution [46]. This further enables blockchain to support the reliable operation of EHR management and medical auditing systems [47].

In recent studies, blockchain has been used as a reliable platform for tracing patients' medical histories [48], and it has been shown that blockchain-based EHR sharing platforms can provide high security and scalability [49]. Meanwhile, blockchain has been widely applied to the design of audit models for medical data [50]. Further, using smart contracts to implement a transparent and automated auditing mechanism [51] ensures the reliability of both input and output data, thus enhancing the verifiability and analyzability of EHR [52].

Recent progress in blockchain applications has been one of the motivations for this study. Namely, blockchain can serve as a reliable and scalable platform for managing personal EHRs [53], ensuring efficient data integration and traceability and thus helping in developing a personalized RAG mechanism by providing

high-quality data for RAG. In addition, blockchain can act as a regulatory platform for LLM outputs [54], thus ensuring the authenticity, reliability, and traceability of a model's output through blockchain-based audit mechanisms and on-chain records. This further ensures transparency and compliance of the model's output and has the potential for establishing high levels of trust and accountability in artificial intelligence applications [55].

III. PROPOSED SCHEME

The main contribution of this work is the proposal of the CHATCBD prototype, which represents a blockchain-integrated, chat-based digital hospital. This section introduces the operational logic of the CHATCBD, as shown in Fig. 2).

A. Consulting System

The main goal of the consulting system is to conduct precise triage and registration so that CHATCBD can adapt to the unique healthcare needs of a diverse patient population and patients can be directed to the most appropriate department for their specific treatment needs. Our consulting system's design was inspired by hospital workflows, and it enhances the efficiency and accuracy of agent diagnoses in the next phase. This system involves the following key components:

- *Assistant Nurse*: An agent is primarily engaged in initial patient interactions, analyzing a user's basic input to determine whether further detailed information from the user is required.
- *Triage Nurse*: This agent works in unison with the assistant nurse to formulate a strategy and offer dependable advice on suitable departmental referrals for the patients.
- *Head Nurse*: The head nurse assumes the crucial function of making conclusive decisions regarding patient information, ensuring that patients are accurately triaged to the relevant departments based on their symptoms.
- *Blockchain*: A secure blockchain ledger stores and manages patient data as well as encompasses information from numerous prior users of CHATCBD [56], providing high-quality data for the Head Nurse's RAG.

Figs. 3 and 4 illustrate that the structured interaction processes are divided into three interconnected stages. The first stage, which is primarily featured in the figure, is centered around the interaction between the patient and the assistant nurse. Here, the assistant nurse's role is to gather fundamental patient data such as gender, age, height, weight, and presenting symptoms. This crucial data collection is mathematically represented as set I (where $I = i_1, i_2, \dots, i_n$) and forms the empirical groundwork for all subsequent decision-making processes in the healthcare pathway. Following this initial data gathering, the second stage commences, which involves a collaboration between an assistant nurse and a triage nurse. A triage nurse plays a vital role here as it acts as an intermediary and specialist in symptomatic triage. Specifically, a triage nurse is empowered to filter and refine these suggestions to align with the most pertinent departmental services.

As depicted in Fig. 4, in the subsequent phase, the head nurse conducts a detailed examination of the departmental allocations suggested by the short-term conversational context memory. This process is also facilitated by the blockchain



Fig. 2. A digital hospital integrating LLMs with blockchain.

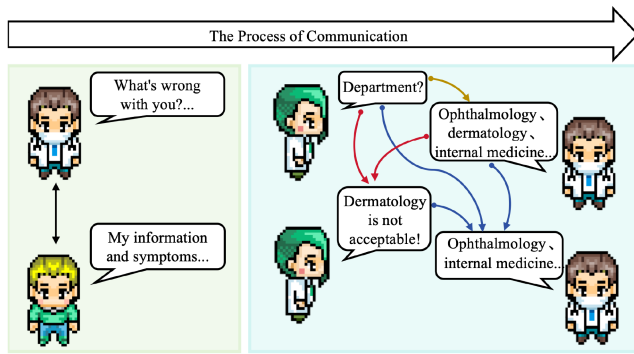


Fig. 3. Illustration of interactions between a patient and an assistant nurse and between an assistant nurse and a triage nurse.

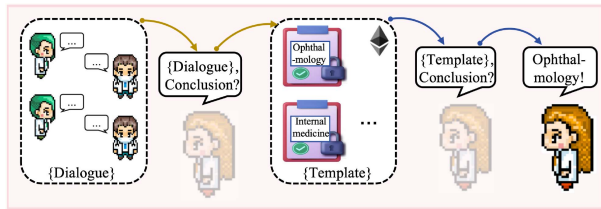


Fig. 4. Blockchain-based conclusion.

framework, where patient templates are categorized and stored in smart contracts by department [57]. As such, rather than individually reviewing each department's smart contracts, the head nurse focuses on the filtered recommendations provided by the triage nurse. Accordingly, by leveraging the insights gleaned from this blockchain-anchored data, the head nurse can construct a well-informed and definitive determination of the patient's optimal departmental referral for consultation.

B. Diagnosing System

In the diagnosing system, as shown in Fig. 5, the primary objective is to diagnose patients and produce diagnostic results and treatment recommendations. This system involves the following key components:

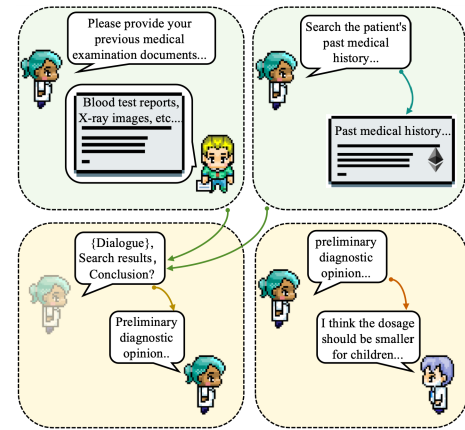


Fig. 5. Diagnosing process.

- *Associate Chief Doctor*: This agent initiates patient evaluation by requesting specific medical documents (e.g., recent blood test reports or X-ray images) represented by a set D , where $D = \{d_1, d_2, \dots, d_n\}$. These documents provide crucial information on a patient's current health status and help form a preliminary diagnostic opinion.
- *Blockchain*: An associate chief doctor also uses a patient's blockchain address to access his/her past medical records. Blockchain ensures the immutability and traceability of a patient's EHR, assuming the patient has used this framework previously.
- *Chief Doctor*: The chief doctor finalizes the diagnostic report, adding any missing data and refining it to ensure accurate treatment recommendations tailored to the patient's specific needs.

In summary, an associate chief doctor compiles a preliminary medical report tailored to meet the individualized health needs of a patient based on information gathered from the patient's input and blockchain records. This report includes an assessment of the condition and corresponding treatment recommendations. Once drafted, the report is passed on to a chief doctor for refinement.

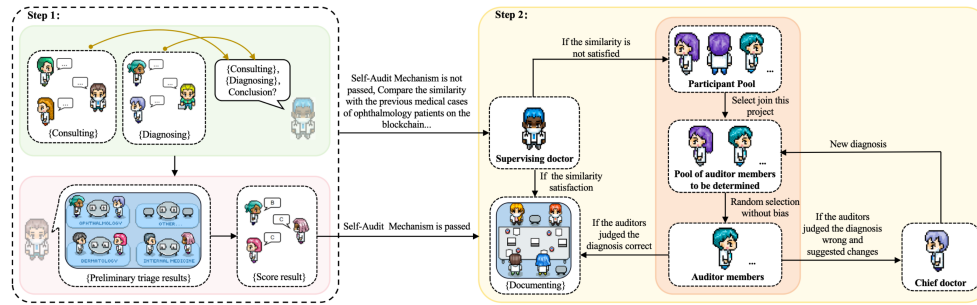


Fig. 6. Auditing process, where pink solid-line boxes indicate steps directly handled by human audits.

C. Auditing System

The auditing system serves as a safeguard to ensure the reliability of the LLMs' outputs, addressing the potential issues of unreliability, as shown in Fig. 6. Although the integration of multiple agents and the application of RAG to the first two systems can reduce errors, hallucinations in the output can still appear. From the perspective of a hospital's practical operations, it is not recommended to rely only on medical automation technologies to provide definitive results. Therefore, it is essential to ensure that only verified and audited content is permitted to proceed to the documenting system. This system involves the following key components:

- **Supervising Doctor:** This agent is responsible for convening a multidisciplinary team of doctors, based on the summary provided by a triage nurse, to audit the diagnosis results. It is also responsible for blockchain-based RAG, and it compares new cases with the existing templates in a blockchain to determine whether a direct progression to documentation is appropriate or if a human-audited process is needed.
- **Chief Doctor:** In scenarios requiring auditing, the chief doctor reviews the outputs post-audit to ensure that the final diagnosis is sound and ready for documenting.
- **Participant Pool:** A cohort of registered auditors within the auditing phase contract who are motivated by the prospect of earning remuneration through their participation in the auditing process. They stand ready for selection to review and validate projects within the framework.
- **Pool of Auditor Members to be Determined:** This group comprises individuals from the participant pool who have expressed a willingness to engage in the audit of a particular project.
- **Audit Members:** These are the blockchain users from the pool of auditor members to be determined who are selected to audit the LLM outputs.
- **Blockchain:** Blockchain safeguards the immutability and traceability of medical records and autonomously enforces a mathematical model for fair and honest auditing.

This system needs to provide an audit of the LLMs' outputs from the diagnosing system. Step one involves the LLMs conducting a self-audit, with the details of the algorithm provided in Section IV-A. A supervising doctor will convene relevant specialists, identified by a triage nurse, and invite them to review the diagnosis results generated by the diagnosing system. If the

self-audit mechanism is not passed, the system will proceed to Step 2, incorporating blockchain technology. The supervising doctor takes on the task of performing blockchain-based audits. This is a process of RAG, which involves retrieving the most relevant cases from the existing templates stored in the blockchain and comparing their similarity with the new cases. If a new case closely resembles a previous case, it is directly forwarded to the documenting phase to avoid redundant audits and expedite patient care. However, when there is no significant similarity match, the framework calls upon the collective intelligence of human auditors. If a majority concludes that the diagnosis is correct, the process transitions seamlessly into the documenting phase. Conversely, if the output is deemed unreliable, it is flagged for further review. In this case, the committee members who identified discrepancies are required to suggest corrections. The chief doctor collects these suggestions and generates a revised outcome, which is then re-submitted to the audit committee for another round of review. Details about the mathematical model for auditing are provided in Section IV-B, Blockchain-based Audit Framework.

D. Documenting System

Four agents in the documenting system are used to perform two main tasks. The first task is to summarize the key outcomes of the first three systems. If the content has been reviewed by Audit Members, it is converted into an IPFS index using the Web3 system, and the index is stored in the smart contract. However, if the content has not been reviewed by Audit Members, its consistency, rigor, and logical coherence are ensured. The second task is to consider which previous cases might be relevant to the current diagnosis, and to record this reasoning by defining an array index in the smart contract [58], ensuring that relevant information is readily accessible for future personalized treatment decisions. This system involves the following key components:

- **Personalized Treatment Plan Organization:** This task is assigned to two doctor agents. If a patient's treatment plan has recently passed auditing systems by blockchain users, the plan is directly provided to the patient. In cases involving audited response templates, as is shown in Fig. 7(a), one doctor is responsible for compiling and integrating this information to maintain consistency while the other reviews and finalizes the document.
- **Creation and Blockchain Uploading of Structured Medical Records:** As shown in Fig. 7(b), a doctor is tasked with

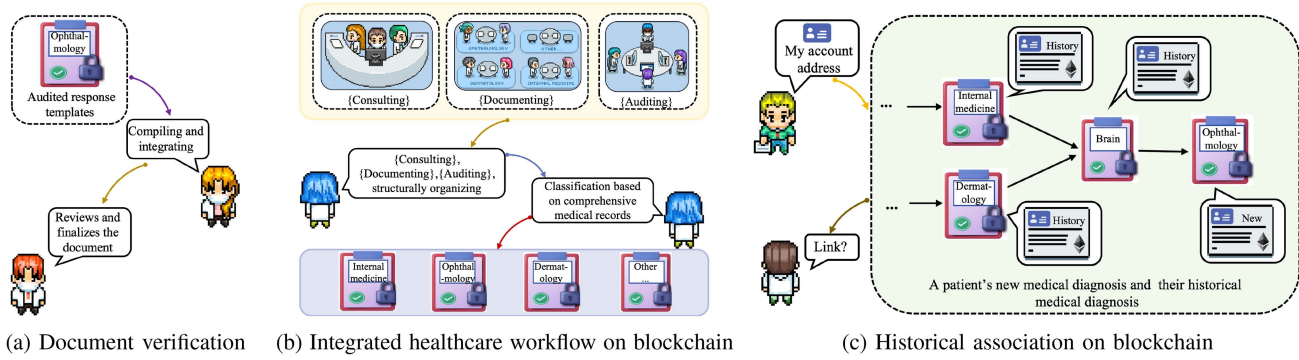


Fig. 7. Documenting process

structurally organizing the detailed records obtained from the first three systems to form comprehensive EHR. These records are subsequently uploaded to the blockchain and are categorized according to the medical department.

- **Linking Current and Historical Health Narratives:** As shown in Fig. 7(c), the fourth doctor's role involves linking a patient's new case with their historical cases stored on the blockchain. This step not only facilitates future indexing and fast access to cases but also provides agents with a form of long-term memory on the blockchain to aid future decision-making.

Through this approach, as shown in Fig. 7. CHATCBD ensures that its medical framework is not only efficient and reliable in the production and auditing of treatment plans but also demonstrates efficiency and systematic organization in case management and historical record integration.

IV. KEY TECHNIQUES

CHATCBD's core is made up of two main sections. Internally, multi-agent role-playing is used to achieve consensus across four systems driven by LLMs. Externally, a blockchain-based memory and audit framework are used to provide a reliable reasoning foundation and management framework.

A. Multi-Agent Collaboration With Internal Enhancement

1) **Role Assignment:** This study delves into the sophisticated role-playing capabilities of the LLMs applied in the CHATCBD digital healthcare system. The proposed system is structured into four distinct operational phases, namely consulting, diagnosing, auditing, and documenting phases, each of which leverages the LLMs as specialized agents programmed to adjust their responses to align seamlessly with their designated roles and interaction contexts.

- **Consulting Phase:** In this phase, LLMs serve as digital nurses handling initial patient interactions with a focus on empathy and data collection.
- **Diagnosing Phase:** In this phase, LLMs act as digital doctors who analyze patient data and formulate preliminary diagnoses.
- **Auditing Phase:** LLMs in this phase ensure the reliability and consistency of the diagnostic process.
- **Documenting Phase:** In this final phase, LLMs meticulously organize and secure medical records.

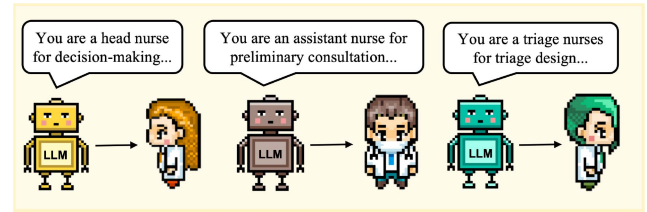


Fig. 8. Role Specialization in CHATCBD.

The use of role-specific prompts is pivotal in predefining the responsibilities of each agent and ensuring that their response style and content are aptly suited to their roles. For instance, in the consulting phase, the head nurse's prompt is denoted by P_H , the assistant nurse's prompt is labeled as P_A , and the triage nurse's prompt is denoted by P_T . These prompts are crucial for setting the roles before any dialogue begins, represented as L_H , L_A , and L_T . The roles are assigned as follows: $H \leftarrow L_H^{P_H}$, $A \leftarrow L_A^{P_A}$, and $T \leftarrow L_T^{P_T}$, as shown in Fig. 8). LLMs simulate the decision-making process of medical professionals and provide scalable and realistic solutions for healthcare settings. Agents work collaboratively in the system and are guided by established communication protocols. The integration of these capabilities demonstrates the potential of LLMs in digital healthcare [59].

2) **Self-Audit Mechanism:** This study develops a self-audit mechanism. Since a preliminary medical diagnosis has been obtained from the diagnosing system, the algorithm proceeds to the auditing system. In the auditing system, Step 1 is as follows: a self-audit mechanism is used to optimize the auditing system. The goal of Step 1 is to gather LLM experts to audit the diagnosing system's result. The audit process consists of specialists assigning ratings to a particular diagnosis using three levels (A - high confidence, B - moderate, C - low confidence). If level A ratings account for 50% or more, the diagnosis proceeds directly to the documenting system; if level C ratings reach 50% or more, the doctor responsible for the diagnosing system must generate a new diagnosis; if both level A and level C ratings remain below 50% or if after two iterations level C ratings exceed 50%, the process advances to Step 2. In simple terms, when the majority of experts consider the diagnosing system's answer to be correct, it directly enters the documenting system; otherwise, further auditing is required.

The fundamental principle of this mechanism is that GPT models generate one token (a word or character) at each time

Algorithm 1: Self-Audit Mechanism.**Inputs:**

- Chief doctors from different fields, $CD = \{cd_1, \dots, cd_n\}$
- Diagnosing chief doctor, cd_0
- Initial medical diagnosis from cd_0 , MD_0
- Maximum iteration number, t
- Prompts for rating and re-diagnosis, $\{p_{rating}, p_{re diagnosis}\}$

Output:

- Final medical diagnosis, MD_f

```

1:  $MD_{cur} \leftarrow MD_0$ 
2:  $n_{iter} \leftarrow 0$ 
3: while True do
4:    $n_{iter} \leftarrow n_{iter} + 1$ 
5:   Ratings  $\leftarrow \emptyset$ 
6:   for each  $cd_i$  in CD do
7:      $rating_i \leftarrow cd_i$  evaluates  $MD_{cur}$  using  $p_{rating}$ 
8:     Add  $rating_i$  to Ratings
9:   end for
10:  Compute  $pct_A$ , the percentage of level A ratings
11:  Compute  $pct_C$ , the percentage of level C ratings
12:  if  $pct_A \geq 50\%$  then
13:    Proceed to the documenting system
14:    break
15:  else if  $pct_C \geq 50\%$  then
16:    if  $n_{iter} \geq t$  then
17:      Proceed to Step 2 (blockchain technology)
18:      break
19:    else
20:       $MD_{cur} \leftarrow cd_0$  generates new diagnosis using
         $p_{re diagnosis}$ 
21:    end if
22:  else
23:    if  $n_{iter} \geq t$  then
24:      Proceed to Step 2 (blockchain technology)
25:      break
26:    else
27:      Continue to the next iteration
28:    end if
29:  end if
30: end while
31: return  $MD_f \leftarrow MD_{cur}$ 

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step and progressively construct the output sequence. Each token is generated based on a probability distribution, where the model assigns a certain probability to each possible token. This probability reflects the model's confidence, indicating how likely a particular token is to be the best choice. Multiple tokens are combined to create a complete output sequence. By generating multiple responses and comparing their consistency, the model's confidence can be assessed [60].

B. Blockchain-Enabled External Regulation

1) Blockchain-Based Personalized Routing Management Mechanism: This study introduces a blockchain-based memory architecture, namely a multi-department blockchain-based personalized routing management mechanism driven by LLMs, as illustrated in Fig. 9. Traditional applications of LLMs

to healthcare often result in hasty diagnoses due to insufficient inquiry and the absence of intermediary steps. In part (b), the CHATCBD presents a more complex and structured process. The implementation of blockchain technology into the CHATCBD system ensures that the CHATCBD output is not only dependent on the original model's decisions but instead integrates the collective wisdom of a decentralized network.

Smart contracts serve not only as a repository for storing patient data, which ensures that the data collected during the consultation and diagnosis phases are transparent, immutable, and traceable [61], but also play a crucial role in combining short-term conversational context memory between the patients and agents, as well as between the agents themselves. However, long-term memory is stored on a blockchain. This combination, which functions as a more personalized and reliable content-relevant prompt, helps in reliable RAG, guiding each phase toward more accurate objectives. When using the CHATCBD, in the consulting system, the Head Nurse routes to the smart contract content of the department summarized by a triage nurse to construct the RAG. In the diagnosing system, an associate chief doctor and a chief doctor make further diagnoses based on the historical EHR of a patient stored in a blockchain. In the auditing system, the supervising doctor routes to the previous diagnostic results from the relevant department to construct the RAG.

After completing the treatment, the system employs blockchain technology to link new patient cases with their historical records. This facilitates the retrieval enhancement from an external knowledge base, providing a means of accessing long-term memory for future personalized decision-making processes. In the blockchain, data are stored as different smart contracts according to the medical department; these smart contracts can also store information about associated medical records linked to a specific patient's address.

2) Blockchain-Based Audit Framework: In the diagnosing phase, chief and associate chief doctors collaborate to devise "Solution A." However, this solution is not directly presented to patients due to potential diagnostic biases, inaccuracies, or inconsistencies with similar cases. To overcome these problems, patient input datasets (denoted by I and D) and Solution A are jointly labeled as "a" and stored in a smart contract. The same storage method is used for the other users' inputs (e.g., b and c) and their outputs (B and C) during the consulting and diagnosing phases.

During the auditing phase, the supervising doctors oversee the process by referencing the smart contract data, such as sets a and A or b and B. For instance, in the ophthalmology field, a supervising doctor compares the current case with previously audited cases stored in a blockchain. If a matching case is identified, suggesting the availability of a pre-audited template for a similar scenario, the process directly advances to the documenting phase. If there is no match, using the patient's consent, a deeper audit is initiated. This process involves distributed collective intelligence. Online blockchain users in the participant pool, who are acting as auditors, proactively select sets a and A from the smart contract based on their expertise to prepare for auditing. The users who select the same set form a candidate committee and wait to be drawn into an actual audit committee. If a patient decides to enter this audit stage, it is assumed that they inherently trust the identity of the auditor. The process of selecting audit members from a candidate committee represents

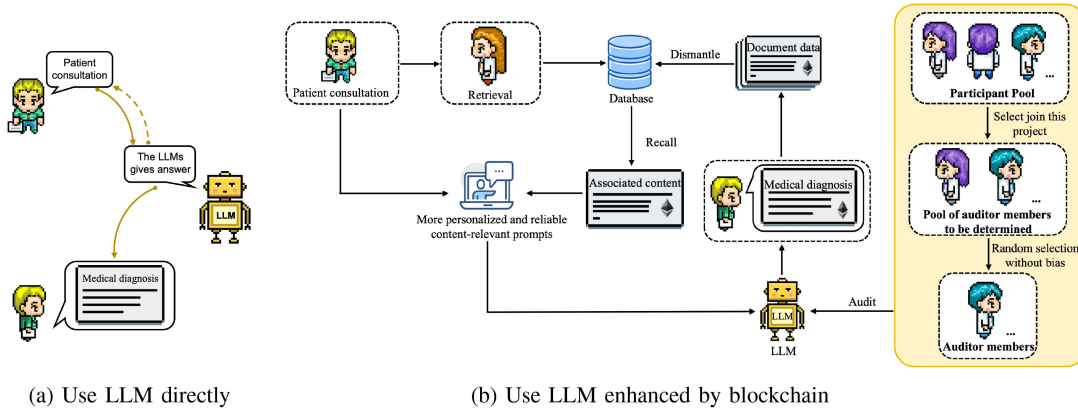


Fig. 9. Medical Diagnostics: Traditional LLM versus the proposed CHATCBD

Algorithm 2: Random Selection Method Without Bias.

Inputs:

- Candidate committee address list, CC , a list of addresses;
- The number of addresses in the list, $len(CC)$;
- The number of online candidates, oc ;
- Required number of members for an audit committee, N ;
- The hash value of a block L at the time of request, l ;
- A character encoding derived number, K ;
- The address of an LLM provider, $p.address$;
- The address of a patient, $pat.address$.

Output:

- Selected audit committee set, AC .

```

1:  $currentBlock \leftarrow L$ 
2:  $seed \leftarrow l$ 
3:  $AC \leftarrow \emptyset$ 
4:  $j \leftarrow 0$ 
5: if  $oc < 10 \times N$  then
6:   throw Insufficient online candidates
7: end if
8: while  $j < N$  and  $currentBlock \leq L + K$  do
9:    $index \leftarrow seed \bmod len(CC)$ 
10:  if  $CC[index].state == \text{Online}$  then
11:    and  $CC[index].address \neq pat.address$ 
12:    and  $CC[index].address \neq p.address$ 
13:     $CC[index].state \leftarrow \text{Auditor}$ 
14:     $oc \leftarrow oc - 1$ 
15:    add  $CC[index]$  to  $AC$ 
16:     $j \leftarrow j + 1$ 
17:  end if
18:   $currentBlock \leftarrow currentBlock + 1$ 
19:   $seed \leftarrow hash(seed)$ 
20: end while
21: return  $AC$ 

```

a random selection process without bias, indicating that neither the providers of an LLM nor the doctors or patients can have advantages in committee selection.

This study develops a random selection algorithm implemented on Ethereum for the selection of the audit committee, as presented in Algorithm 2. First, the auditing phase contract administers the participant pool, candidate committee, and audit committee and provides the necessary interfaces for any blockchain user to register into the participant pool as a node of

reliable distributed intelligence. These nodes aim to earn auditing revenue by examining the LLM output, and users from the participant pool register directly into the candidate committee by selecting a set of consultations they are proficient in when a batch of input-output pairs from LLMs are displayed in the contract window. Further, addresses in the candidate committee are arranged in a list according to the registration order. After registering, users in the candidate committee can toggle their status between online and offline to indicate their availability for selection. Two interfaces are designed in the smart contract for selecting N audit members from the candidate committee to form an audit committee. The request interface is first invoked by a smart contract on a block L . This implies that the transaction involves the contract at a block L 's height, where the hash value of the block is denoted by l . At this point, the value of a number K must be calculated based on the character encoding of the audit tasks a and A . After the blockchain generates K blocks, that is, at a block $(L + K)$, another select interface of the auditing phase contract can be invoked to select N online audit members.

The audit committee does not exceed N members, where the value of N is predetermined. The higher the N value is, the more reliable the LLM output will be, but this will also increase the auditing cost. During the auditing phase, a smart contract calculates a reporting window based on the gas fees consumed by a and A stored on the blockchain. Within this window, the contract accepts reports from the audit committee members. If at least M out of N reports, where M is the majority of N , identify errors in the LLMs' output, a violation is automatically confirmed. Auditors voting for the violation suggest corrections to the interaction layer of the smart contract. All voting results and suggested corrections are stored on the blockchain and publicly available, which ensures transparent sharing between auditors during the auditing phase and is beneficial to later evaluation of auditor behavior. After receiving suggested corrections, the LLM outputs a new result for continued voting by the audit committee. Notably, the smart contract is designed to accept only one report per auditor in the same reporting window.

In a sense, N independent auditors constitute a game with N participants, where each auditor seeks to maximize its own benefit. Throughout the life cycle of blockchain users as auditors [62], an LLM audit layer's smart contract maintains the process. Adopting the principle of Nash equilibrium from the game theory, this study demonstrates that auditors must be

honest participants in a game, meaning that they must vote based on the actual events regarding the LLMs' outputs. In game theory, a typical scenario is the strategic interaction of n players, which is often represented in a matrix form. This definition encompasses the group of players, the configuration of strategies, and the design of the payment function. Briefly, this framework employs mathematical and game theory principles to provide a structured platform for auditors, thus ensuring honest voting on the LLMs' outputs. The definition of the audit game is provided below.

Definition 1. Audit Game: This is an n -player game expressed as a triple (SW, Σ, Π) , where

- 1) $SW = \{a_1, a_2, \dots, a_n\}$ represents a set of n players, each of which is a selected auditor, and together the players form an audit committee;
- 2) $\Sigma = \Sigma_1 \times \Sigma_2 \times \dots \times \Sigma_n$ represents a set of strategy profiles, where Σ_k denotes a set of actions for a researcher r_k , that is, r_k can select any action $\sigma_k \in \Sigma_k$. Thus, a strategy profile represents a vector $\sigma^* = (\sigma_1^*, \sigma_2^*, \dots, \sigma_n^*)$, where σ_k^* is a particular action of Σ_k , ($k = 1, 2, \dots, n$);
- 3) $\Pi = \{\pi_1, \pi_2, \dots, \pi_n\}$ represents a set of payoff functions, where $\pi_k : \Sigma \rightarrow R$ is the payoff function deciding the revenue for a researcher a_k under a certain strategy ($k = 1, 2, \dots, n$); R is the corresponding revenue.

Moreover, $\sigma_{-k} = \{\sigma_1, \sigma_2, \dots, \sigma_{k-1}, \sigma_{k+1}, \sigma_n\}$ is defined as a strategy profile σ excluding player k 's action. Hence, the full strategy can be expressed by $\sigma = \{\sigma_k, \sigma_{-k}\}$. Each auditor has two potential strategies, namely to either report or not report violations, validating a doctor's interaction with a specific medical record, which is expressed as $\Sigma_k = \{\sigma_k^r, \sigma_k^s\}$, σ_k^r and represents reporting violations to the smart contract; σ_k^s indicates the strategy of not reporting violations and maintaining silence toward the smart contract. The violation confirmation is defined as follows.

Definition 2 : Violation Confirmation: Depending on the outcome of a strategy profile in an N -auditor game, the service violation is affirmed only when $\|A_{reporter}\| \geq M$, where $1 < N/2 < M < N$, $N, M \in N$; otherwise, it is assumed that no violation has occurred.

If the number of auditors reporting a violation is greater than or equal to M , the auditors who have reported the violation earn a gain of 10, whereas those who have not reported the violation obtain no gain. Conversely, if the number of auditors reporting a violation is less than M , the auditors reporting the violation receive a negative gain of -1, whereas the non-reporting auditors obtain a gain of one. The first Nash equilibrium occurs when all auditors report a violation. In this equilibrium, the auditors' payoffs are maximized, which means they have no incentive to change their strategy. The second Nash equilibrium occurs when all auditors do not report a non-existing violation. In this case, even though auditors have no payoff, they avoid negative payoffs, which also means they have no motivation to alter their strategy. Thus, an auditor's optimal strategy is to report a violation when there is a violation and not to report a violation when there is no violation. The audit framework concludes under the following condition: fewer than M auditors deem the output of the LLMs as erroneous. In such a case, the LLMs' outputs are considered to possess a certain degree of reliability, thus warranting progression to the documenting phase.

V. EXPERIMENTAL RESULTS

This study conducted a comprehensive experimental evaluation of the four systems of the proposed CHATCBD into three phases. First, without incorporating blockchain technology, 10 healthcare-related question-answer datasets were used to validate the effectiveness of the proposed agent architecture. Next, blockchain technology was applied, and complex, personalized healthcare scenarios from three personalized datasets were selected to assess the optimization effect of the original model's performance in personalized healthcare. Finally, the operational cost of the proposed CHATCBD was evaluated by simulating the deployment of smart contracts in a blockchain environment and testing the system overhead introduced by the integration of blockchain with the agents.

A. Evaluation Method

To evaluate the multi-agent collaborative consensus architecture of the CHATCBD and demonstrate its enhanced reliability and ability to reduce hallucination significantly, this study used accuracy as an evaluation metric. Three open-source datasets, namely the MedQA, MedMCQA, and PubMedQA datasets, were selected, along with the Nanjing Drum Tower Hospital Q&A dataset (NDTHQA).

To evaluate the impact of integrating blockchain, this study conducted tests in two parts. First, the effect of the CHATCBD architecture on accuracy and consistency in personalized healthcare after blockchain integration was evaluated. Since different underlying causes might have similar symptoms, to validate the personalized treatment capabilities of the proposed solution, this study analyzed the accuracy of the LLM using the Fever of Unknown Origin Evaluation Dataset (FUO_ED) [63], the Stomach Pain of Unknown Origin Evaluation Dataset (SPUO_ED), and the Headache of Unknown Origin Evaluation Dataset (HUO_ED). Second, the cost of deploying smart contracts and decentralized intelligent audits after blockchain integration was assessed.

B. Dataset Construction

1) **NDTHQA:** The NDTHQA dataset was constructed to validate the proposed agent architecture. The NDTHQA was derived from clinical diagnostic data and patient treatment logs (in an image or PDF format) at Nanjing Drum Tower Hospital in Nanjing, Jiangsu Province, China. When constructing the NDTHQA dataset, first, Optical Character Recognition (OCR) technology was employed to extract textual information from the de-identified logs. Next, the GPT-4 model was used to analyze, summarize, and annotate the extracted data, which allowed for constructing multiple-choice questions covering seven clinical scenarios, where various clinical indicators of patients were used to form questions, and the diagnostic conclusions for each scenario denoted the answers. After manual validation and screening, 710 high-quality data entries were selected to construct an evaluation dataset that could authentically reflect the diverse scenarios encountered in the hospital. The open-sourced examples were provided from each dataset across seven scenarios.

2) **FUO_ED, SPUO_ED and HUO_ED:** The FUO_ED, SPUO_ED, and HUO_ED datasets were designed to validate the personalized reasoning ability of ChatCBD. The FUO_ED

dataset was derived from the clinical diagnostic data of FUO patients treated at the. First, 510 multiple-choice questions from the FUO_ED dataset, which were applicable to personalized healthcare, were selected to form a test dataset. Then, GPT-4 was used to create 10 patients, each with a series of EHRs. By refining the correct answers in the FUO_ED dataset based on their EHRs, they were made more relevant and reasonable. This led to the creation of a personalized multiple-choice dataset with patient symptoms as inputs and diagnostic suggestions as answers. The SPUO_ED and HUO_ED datasets were initially constructed using GPT-4 to extract data from the MedQA and the NDTHQA dataset, and each of these two datasets contained 510 multiple-choice questions. Then, the same method as before was applied to refine the questions and answers, making them more complex. Some examples from the above dataset have been made open-source.

3) Blockchain Knowledge Base Dataset: The blockchain knowledge base dataset was designed for the proposed RAG mechanism. Up to 1 GB of high-quality doctor-patient dialogue data were collected from the Internet, and then GPT-4 was used to precisely categorize the collected data based on the medical department. The EHRs of the 10 patients mentioned in the previous section were also included. This yielded the construction of the proposed multi-department blockchain knowledge base dataset, which was stored in the IPFS with department-specific indexing, and the IPFS index was stored in the medical record template contract. Some examples from this dataset have been made open-source.

C. Experimental Setup

The software used in the experiments included a Web3 system. As for the front-end side, a user-friendly testing interface was developed using Vue.js, which facilitated interaction with the blockchain; the back-end side was powered by Ganache, MetaMask was integrated for wallet management, and Remix was used for smart contract testing and deployment. The Ganache private chain was used to provide a local Ethereum development environment that could simulate the actual Ethereum Mainnet. This study used a programming language supported by Ethereum, Solidity, to develop smart contracts. In addition, the Ganache was configured with multiple digital accounts, each pre-funded with 100 ETH, thus creating an authentic and controlled environment for testing the CHATCBD system.

Regarding the language model operations in the CHATCBD, OpenAI's GPT-3.5 and GPT-4 served as the core engines. The model parameters were adjusted to achieve optimal performance. In particular, the temperature and top_p were both set to one for all generation tasks. A thorough evaluation was conducted using different prompt methods, including zero-shot, zero-shot with a chain of thought (CoT), few-shot, and CoT with few-shot.

D. CHATCBD Agent Architecture Performance Evaluation

Next, it is evaluated whether the CHATCBD multi-agent architecture could improve the reliability of the LLMs without relying on blockchain-based knowledge summarization and decentralized intelligence auditing. For this purpose, this study used role-playing, task division, and consensus-based diagnostic processes. In the consulting system, the head nurse no longer

performed triage reasoning based on the case database managed by blockchain but directly generated the most likely triage department. In the diagnosing system, the associate chief doctor skipped the reasoning that considered both the blockchain and the potentially relevant medical history. In the auditing system, Step 2 was skipped, and the final result of Step 1 was directly taken as the final answer of the CHATCBD. Experiments were conducted on three public datasets and seven real medical scenario datasets, including Surgical Strategy, Clinical Knowledge, Postoperative Care Knowledge, Chronic Disease Management, Patient Communication Skills, Preventive Medicine, and Emergency Knowledge datasets. As shown in Table I, the evaluation results of GPT-3.5 and GPT-4 on various medical tasks were compared using different methods, including the CHATCBD architecture. The results indicated that the proposed CHATCBD architecture achieved the highest accuracy among all methods, which demonstrated its significant application potential in the medical field. Namely, it outperformed traditional few-shot and CoT methods and could effectively mitigate hallucinations.

E. Blockchain and LLM Integration Performance Analysis

This study conducted additional tests to examine the impact of the CHATCBD architecture on the reliability and consistency of LLMs in personalized diagnosis. To this end, professional doctors were invited to participate in the study as online blockchain auditors. On the FUO_ED, SPUO_ED, and HUO_ED datasets, GPT-3.5 and GPT-4 were used, and each of them employed different prompt strategies for experimentation. The experimental results clearly showed that the decentralized intelligent architecture of the CHATCBD, using an external blockchain, performed better in terms of diagnostic accuracy. The CHATCBD could effectively classify and grade certain personalized diagnostic problems that lacked reliable template content and pass them to decentralized intelligence for further auditing. Errors in the LLM's responses were treated as valid experiences, stored on the blockchain as long-term memory, and were expected to be recognized in the next round of dialogue. This allowed the audited long-term memory on the blockchain to contribute to more personalized and accurate answers. This also confirmed that the CHATCBD architecture could be well-suited for real-world medical scenarios with personalized and reliable requirements.

There could be two main reasons for the CHATCBD's high accuracy. First, upon recognizing that blockchain could naturally serve as a scalable foundation for a RAG-based EHR knowledge base, this study developed a blockchain-based personalized routing management mechanism. The knowledge base data of each department and the test data from the MedQA were projected into the vector space using the trained medical embedding model. The results are presented in Fig. 10, where the gray represents the test data, and the other colors denote the datasets from different departments. As shown in Fig. 10, there was a significant overlap between the departmental datasets, which indicated semantic similarity in the vector space. Since the self-developed knowledge base dataset was as large as 1 GB, this overlap could easily lead to substantial data conflicts. This was why, under the traditional RAG mechanism, it was difficult to recall high-quality data. Conversely, the graded management of the blockchain knowledge base could effectively alleviate data conflicts and enable the recall of relatively high-quality

TABLE I
EVALUATION RESULTS OF THE GPT-3.5 AND GPT-4.5: BOLDED RESULTS INDICATE THE BEST PERFORMANCES

Method	MedQA	MedMCQA	PubMedQA	Surgical strategy	Clinical knowledge	Postoperative care knowledge	Chronic disease management	Patient communication skills	Preventive medicine	Emergency knowledge
<i>*Few-shot setting GPT-3.5</i>										
Few-shot	54.6	56.7	67.6	56.3	70.2	57.3	70.6	79.1	72.1	64.5
Few-shot CoT	55.3	54.7	71.0	49.7	66.9	54.4	58.6	77.1	60.0	58.6
<i>*Zero-shot setting GPT-3.5</i>										
Zero-shot	54.3	56.3	73.7	62.1	75.1	61.7	72.2	84.2	73.4	66.5
Zero-shot CoT	44.3	47.3	61.3	62.5	63.5	54.1	64.9	85.3	64.1	56.7
CHATCBD (Proposed)	63.5	57.1	74.3	69.8	75.7	67.2	73.2	89.3	74.5	68.9
<i>*Few-shot setting GPT-4</i>										
Few-shot	76.6	70.1	73.4	77.1	87.9	74.2	91.8	90.4	89.5	80.9
Few-shot CoT	73.3	63.2	74.9	76.9	88.4	60.1	80.5	88.2	70.6	71.6
<i>*Zero-shot setting GPT-4</i>										
Zero-shot	73.0	69.0	76.2	73.2	82.1	73.8	88.4	88.5	88.4	79.2
Zero-shot CoT	61.8	69.0	71.0	76.5	83.7	78.9	91.3	91.9	90.0	79.7
CHATCBD (Proposed)	81.7	75.1	77.0	77.5	89.4	88.6	94.9	93.0	94.3	83.2

Each column contain 2 bold values, representing the best accuracy for GPT-3.5 driven methods and GPT-4.5 driven methods respectively.

TABLE II
EVALUATION RESULTS OF INTEGRATING BLOCKCHAIN IN GPT-3.5 AND GPT-4.5: BOLDED RESULTS INDICATE THE BEST PERFORMANCES

Method	FUO_ED	SPUO_ED	HUO_ED
<i>*Few-shot setting GPT-3.5</i>			
Few-shot	7.4	18.1	16.9
Few-shot + RAG	10.0	24.6	23.1
Few-shot CoT	7.9	18.4	17.0
<i>*Zero-shot setting GPT-3.5</i>			
Zero-shot	8.2	19.0	18.3
Zero-shot CoT	8.6	21.3	20.6
CHATCBD (Proposed)	62.7	64.0	61.9
<i>*Few-shot setting GPT-4</i>			
Few-shot	42.6	53.2	51.7
Few-shot + RAG	44.0	55.0	58.2
Few-shot CoT	40.5	52.0	54.8
<i>*Zero-shot setting GPT-4</i>			
Zero-shot	41.3	52.7	50.1
Zero-shot CoT	39.9	50.6	55.8
CHATCBD (Proposed)	83.4	86.0	87.7

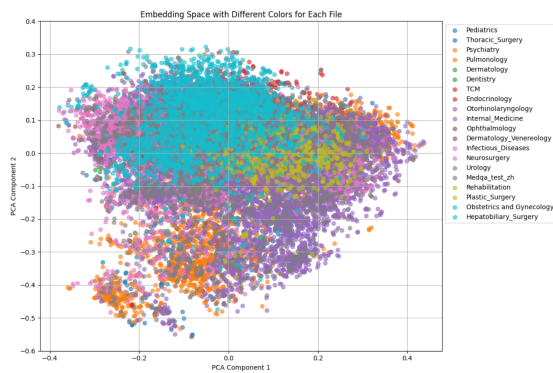


Fig. 10. The vector space projection diagram of the blockchain data of each department and the MedQA test data.

data, thus significantly improving the quality of the model's responses. Further, smart contracts were used to construct the blockchain-based audit framework to provide additional security against the hallucination in the LLMs. In the experiments, over 60% hallucination issues were accurately detected by the proposed LLM self-audit mechanism and subsequently entered the

TABLE III
SMART CONTRACT DEPLOYMENT GAS USAGE AND COST IN ETHER

Contract	Gas used	Cost (Ether)
Consulting phase contract	991,277	0.02379065
Diagnosing phase contract	600,433	0.01441039
Auditing phase contract	716,722	0.01720133
Medical record template contract	892,986	0.021431664

blockchain-based audit framework. The mathematical model of the proposed smart contract ensured that professional blockchain auditors could make expert judgments, reversing the hallucinated outputs. This process was also recorded by the smart contract and could be used for subsequent RAG, thus establishing a feasible ecosystem for hallucination detection and resolution.

F. Blockchain and LLM Integration Cost Analysis

This section analyzes the costs of the blockchain and LLM integration into the proposed CHATCBD healthcare system. The evaluation results of the blockchain deployments for essential processes revealed that the side effects on the system economics and agent performance were modest.

1) *Smart Contract Deployment and Main Cost*: The CHATCBD system used four smart contracts that automated the rules required for the four primary stages, namely the consulting, diagnosing, auditing, and medical record template creation stages. These contracts underpinned the CHATCBD's transparency, auditability, and traceability.

- The consulting phase contract and the diagnosing phase contract documented the interactions and outcomes between the CHATCBD's users and agents;
- The auditing phase contract logged the decisions made by auditor members and enforced the rules of arbitration automatically;
- The medical record template contract was used for storing patients' records within a specific medical department.

The consulting phase, diagnosing phase, and auditing phase contracts required a single deployment by the developers at the initiation of the CHATCBD project. In contrast, the medical record template contract was deployed multiple times to reflect the division of medical departments (e.g., ophthalmology, internal medicine, and surgery), where each deployment represented a distinct outpatient department. As shown in Table III, the

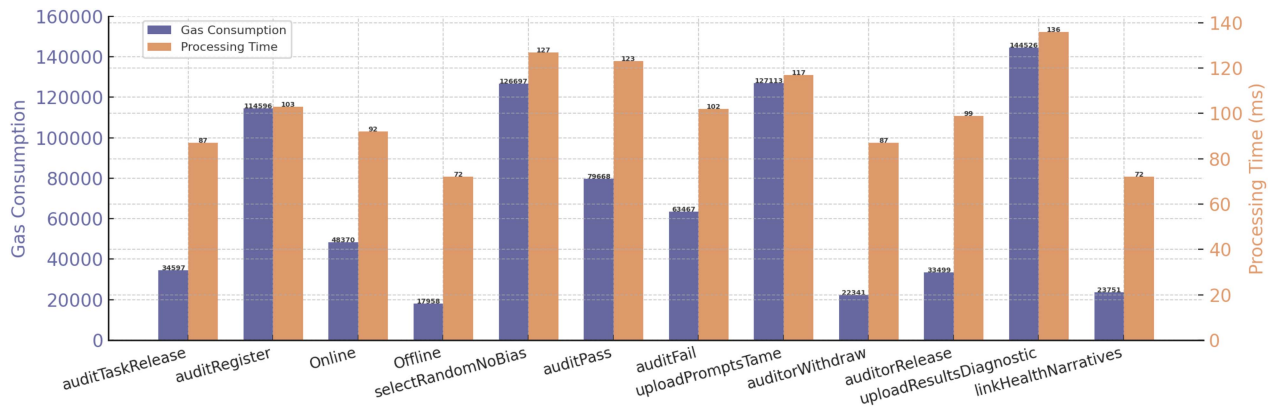


Fig. 11. The gas consumption and processing time of the key functions in smart contracts.

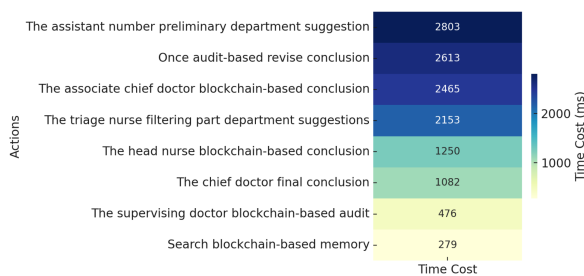


Fig. 12. Time cost (in ms) of agent actions.

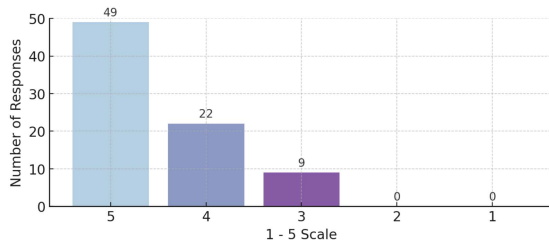


Fig. 13. Survey results: the evaluation of the CHATCBD's reliability and consistency using the five-point scale, where "0" indicates low and "5" denotes high).

deployment of these contracts could accurately replicate the actual operational resource consumption and costs.

To test the proposed model and system design comprehensively, this study invited fifteen doctors from Nanjing Drum Tower Hospital to participate as distributed smart contract actors on the blockchain. The FUO_ED question dataset was used to conduct thorough tests on smart contracts deployed on the Ganache platform. Fig. 10 displays the results of a single use of the CHATCBD, including the gas consumption and transaction processing time associated with key technologies. The more complex the design of critical functions within the smart contracts was, the larger the transaction fees and processing times were. In this analysis, "Gas" denoted a unit representing the computational work miners needed to execute transactions. Transaction fees were calculated by multiplying the amount of gas consumed by the price per unit of gas. The results indicated that the gas cost was primarily concentrated in the auditing phase but remained within an acceptable range. In the blockchain auditing phase, although patients added to

the operational cost, they received more reliable, controllable, auditable, and recyclable LLM diagnostic results. This reflection was crucial for assessing the economic feasibility of implementing blockchain technology in the healthcare sector. In addition, information on gas consumption and time expenditures was obtained by experiments based on the current implementation. However, there is still potential for further gas consumption and time cost reduction through the optimization of the interface implementation, which could be one of the future work directions.

2) Agent Performance Evaluation: Next, this study simulated real-user consultation scenarios to evaluate the CHATCBD system's performance and analyzed the response times of agents at key stages. The response times of agents were related to the output speed of different LLMs and the blockchain feedback time. In this simulation experiment, the blockchain feedback environment was relatively stable, the feedback settings for GPT-4 were fixed, and the input length of patients in the private dataset was similar. Based on this, a conclusion was drawn. The results indicated that the CHATCBD system that combined the LLMs and blockchain technology could effectively handle complex medical consultation requests while maintaining a high level of decision quality. The test results of the agent behavior performance are presented in Fig. 12), where it can be seen that the agents' performances in the consulting, diagnosing, and auditing phases were not affected by the blockchain's throughput. Searching for information on the blockchain took approximately 279 ms each time, and these three stages primarily involved querying related information on the blockchain. The blockchain served as a long-term memory for agents and aided in reflection and summary. Thus, the introduction of blockchain technology not only enhanced the transparency and security of using LLMs but also improved their standardization and auditability.

G. Comprehensive System Assessment

A few factors might influence the CHATCBD users' comprehensive assessment of the system. For instance, the caliber of LLMs can significantly impact user satisfaction; also, the length of the audit phase, which is conducted by human auditors, can also influence user experience. Therefore, to optimize efficiency, this study defined a maximum time limit for auditors. These designs were elaborately outlined in the subsection named Blockchain-based Audit Framework within the Results section of the manuscript.

Although this system has this flaw, the authors believe that its impact on the proposed system's performance is not significant. Namely, an increasing number of studies have indicated that introducing a slow thinking mechanism is necessary to enhance the reliability of LLMs [64]. In addition, even the blockchain-based audit framework inevitably demands additional time. However, based on practical experience, the time spent on appointments, queuing, and waiting for results in an actual hospital process is unavoidable. In contrast, the proposed digital hospital architecture could significantly enhance accessibility to high-quality doctors in the real world.

Cumulatively, the experimental results demonstrated the CHATCBD system's practicality. In the same cohort of 80 participants who had previously assessed GPT-4 in healthcare, this study conducted a follow-up survey focusing on their perceptions of the GPT-4-driven CHATCBD. As illustrated in Fig. 13, the results revealed a pronounced endorsement of the proposed CHATCBD. Notably, the majority of respondents (49 out of 80) attributed a top score of "5," which signified a robust confidence in the proposed system.

VI. CONCLUSION

This study first proposes a medical diagnostic framework that synergistically combines LLMs with blockchain. This innovative approach not only achieves an advancement in the field of massive, personalized, reliable healthcare but also aims to enhance global medical equity. Further, the blockchain-based personalized routing management mechanism is developed, and it can be abstracted as a blockchain knowledge base routing management mechanism. This mechanism, along with the proposed blockchain-based audit framework, is not limited to the application in the digital healthcare industry but can be extended to multiple fields, such as education and law.

Although the proposed CHATCBD framework has shown promising results, there are still a few aspects that could be addressed in future research. First, privacy protection-related problems must be addressed to achieve the widespread application of LLMs in the healthcare domain. Currently, patients face some common concerns when using LLMs, which can be summarized as follows: (1) Is it secure and safe to submit personal data directly to a specific LLM AI company for analysis?; (2) Do users have the right to control and delete their information after submitting their personal data?; (3) What privacy protection measures are allowed during data transmission and storage? Therefore, future work could investigate the development of specialized privacy protection mechanisms, to meet the privacy protection requirements of patients when using LLMs [65]. Finally, regarding the blockchain component, the interface implementation could be further optimized to reduce natural gas consumption and enhance the functionality of smart contracts.

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